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Project Title:	Mechanism for the Crystallization Studies of ZBLAN		
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Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Glasses and ceramics		
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Zip Code:	35812	**Congressional District:**	5
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Key Personnel Changes/Previous PI:	Ethridge, Edwin		
COI Name (Institution):	Tucker, D. (NASA Marshall Space Flight Center) Kaukler, W. (University of Alabama, Huntsville) Antar, B. (University of Tennessee Space Institute)		
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Task Description:	This is a four year study to examine the effects of viscosity, shear, and stress on the crystallization of ZBLAN glasses with the purpose of testing a theory for their crystallization. The non-Newtonian viscosity behavior is being investigated. The study is extended for another year in order to complete the low upmass International Space Station (ISS) experiment "Fluid Merging Viscosity Measurement", FMVM, on ISS Increment 9, 10 and 11. Samples are melted and quenched in 1-g under quiescent conditions at a number of controlled cooling rates to determine crystallization behavior. Experiments are performed on the materials under controlled vibration, shear rate, and stress rate conditions and compared with quiescent experiments. Experiments mayl also be performed in low-g and 2-g conditions on the KC135 aircraft during low-g parabola flights. The viscosity of the glass is measured at different shear conditions and the shear dependence on viscosity is examined. The shear conditions within experimental samples are modeled with computer codes developed for fluid dynamics. Actual experimental conditions are input into the calculations to determine the shear and stress levels within the liquid of the sample under investigation. This information is to used to evaluate the shear dependence on viscosity of ZBLAN in order to determine the possible non-Newtonian viscosity effects on crystallization behavior. New methods for viscosity measurement are being developed for viscosity measurement of viscous liquids as well as the shear rate determination. The objective of this research is to test the hypothesis that shear thinning in liquid ZBLAN contributes to enhanced crystallization of the material and to determine if low-g processing might enhance the glass formability of ZBLAN by greatly reducing fluid flow during undercooling and glass formation. A secondary objective is to investigate non-Newtonian viscosity behavior and methods to determine viscosity at very low shear rates. A number of research teams have observed that glass-forming melts that are solidified in low-g exhibit enhanced glass formation. This project is examining one of these glasses in particular, the heavy metal fluoride glass, ZBLAN. ZBLAN is a potential communication fiber optic glass that has extremely low theoretical signal losses such that information could be transmitted much further than with silica, the current fiber optic material. Much research was performed 15 years ago but ZBLAN never achieved the goals of ultra low signal loss. This was due in part to micro-crystallization in the glass fibers. In this ground based research program, mechanisms for the reduction of crystallization and enhanced glass formation will be examined. If it is possible to develop experimental methods which reduce the crystallization of ZBLAN (perhaps in low gravity) there could be a renewed interest in this technologically significant material and could impact the billion dollar per year fiber optic cable market. The non-Newtonian behavior of glasses is not well characterized. New methods to measure the viscosity at very low shear rates would have technological signifance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The non-Newtonian behavior of glasses is not well characterized. New methods to measure the viscosity at very low shear rates would have technological signifance. The viscosity determination method in the FMVM experiment will permit the measurement of the viscosity of glasses at large undercoolings. Also, the examination of the coalescence of drops has direct application to liquid phase sintering which could become a very useful process for fabricating products from in-situ lunar regolith.
Task Progress:	Low gravity crystallization studies were planned with ZBLAN glass samples. An existing KC135 experiment pallet with data acquisition system and furnace controller was adapted by collaborators at U. Alabama in Huntsville. A 750 watt lamp ellipsoidal furnace was used to heat ZBLAN samples on Platinum thermocouples. A vacuum system at MSFC was adapted with a glove so that it could be pumped to a vaccum, backfilled with inert gas, repeated several times, followed by sample manipulation with the glove. A Pt strip heater was used to heat and melt a chemically cleaned ZBLAN sample. A very fine Pt thermocouple was inserted into the melt to apply a coating to the tip. The samples on the thermocouples were inserted into a vitreous silica tube and sealed in the clean environment. The ellipsoidal furnace control computer was programmed to rapidly ramp the sample temperature up to an isothermal heating temperature, hold for the period of time during the low-g parabola, followed by a rapid quench. This experimental plan approximates isothermal heating of the sample till crystallization. The results were not conclusive. The length of time of low-g parabola was too short to achieve crystallization. The program was quickly re-written to perform multiple heating runs during the low-g parabola. Unfortunately, the amount of heat released during crystallization was too small to detect with the thermocouple. Therefore, it was not possible to determine a time for crystallization for a Time-Temperature-Transformation crystallization boundary. Samples did crystallize in 1-g, at what appears to be shorter times at the crystallization temperature, but could not be compared with the low-g experiments due to the lack of a distinguishable signal with the latter. The conclusion to be drawn from these experiments is that longer low-g experiments appear to be required to perform the experiments without compromised scientific results. Another set of crystallization experiments were performed to examine the effect of shear on crystallization of molten ZBLAN. This experiment was performed in the vacuum glovebox. Due to extreme reactivity of the ZBLAN samples to trace atmospheric water at high temperatures, the surface of the samples reacted and crystallized before the experiment could be completed. Improvements to the vacuum glovebox must be made to achieve useful results. The chamber needs to be made tighter with fewer leaks, and it should be baked out while pumping a vacuum to remove traces of adsorbed water. Work progressed on modifying the Dynamic Mechanical Thermal Analyzer, DMTM, for direct viscosity measurements. Three problems that were addressed are, 1) the thermal loss down the mechanical pushing motor, 2) the thermal expansion of all the structural components in the sample chamber and the expansion of the mechanical pushing mechanism, and 3) heat loss of the bottom of the sample to the mechanical pushing mechanism. An untra low thermal conductivity plate was added to the mechanical pushing mechanism. This greatly limits the heat flow from the sample and the sample chamber into the mechanical pushing mechanism. This helps with sample isothermality and thermal expansion of the mechanical pushing mechanism. The mechanical structure holding the sample was replaced with low thermal expansion components. These adaptations should greatly improve the baseline of the thermal mechanical measurements. The numerical modeling was completed and will be compared with experimental data when available Cylindrical samples have been prepared for the squeeze flow viscosity measurement method. These tests are continuing. In June 2004 the opportunity to propose a "Low Upmass International Space Station Experiment" opened. An experiment was proposed to validate a method for determining the viscosity of viscous liquids by correlating the rate of

	coalescence of 2 liquid spheres. Prior studies had been performed on KC-135. Upon selection, the investigation team redirected our efforts to this high value experiment. An experiment was quickly developed that utilized equipment and supplies on-board the ISS and a small amount of material launched on a Russian Progress rocket. Several flight qualified medical syringes were located in storage at JSC. These were loaded with several low toxicity high viscosity liquids that were degassed in vacuum. The liquids selected were (2 syringes of) glycerin to correlate with our KC-135 experiments, 2 different high viscosity silicone oil viscosity standards, 2 syringes of mineral oil with dissolved poly-isobutylene (a very non-Newtonian viscosity liquid), 2 syringes with honey (one from off the shelf, the other heated to remove water and greatly increase the viscosity), and 2 syringes with corn syrup (one from off the shelf, the other heated to remove water and greatly increase the viscosity). Just before handover and delivery of the 10 syringes, it was observed that the mineral oil with polyisobutylene was reacting with the elastomer syringe plunger seal. These 2 liquids were removed from the shipment to Russia. The samples were launched and arrives at ISS in December 2004, experiments were ready the first of the new year. We waited for the opportunity to have the experiment performed till the summer of 2005. The ISS Science Officer, Mike Fincke, performed the experiments using the Multipurpose Work Area, MWA. Tools in the soldering kit were used to hold a background card with a grid. Nomex thread from the ISS Sewing Kit were used to hold the liquids and allow one liquid drop to be slowly brought into contact with the second liquid. Two equal size liquid drops were deployed onto two strings. Glycerin proved deploy onto the strings. The viscosity was too low and or the surface tension too high. The two silicone oils were deployed easily and 5 coalescence experiments were recorded with the 2 different viscosity liquids. A range of drop sized from 0.5 cc to 4 cc were coalesced. The honey syringes showed extensive crystallization. These were set aside and re-stowed for later experiments. The 2 corn syrup liquids formed a stiff skin Real time discussions with the astronaut saved this experiment. The procedures were changed real time. New drops were deployed rapidly so as to avoid the skin formation. Several successful coalescence runs were obtained with the corn syrup. We are waiting for the opportunity to perform the experiments with the 2 honey liquids after heating the honey in the US Food Warmer. The preliminary results from data analysis indicated a very good agreement with the predicted coalescence time. This indicates that the experiment does validate our model for this new low shear rate viscosity measurement method.
Bibliography Type:	Description: (Last Updated: 10/19/2009)
Articles in Peer-reviewed Journals	Antar BN, Ethridge EC, Maxwell D. "Viscosity measurement using drop coalescence in microgravity." Microgravity Science & Technology. 2003 Mar;14(1):9-19. http://dx.doi.org/10.1007/BF02873332 , Jan-2004